

Question:

An isosceles trapezoid has an inscribed circle tangent to each of its four sides. The radius of the circle is \$3\$, and the area of the trapezoid is \$72\$. Let the parallel sides of the trapezoid have lengths \$r\$ and \$s\$, with \$r \neq s\$. Find \$r^2+s^2\$

Reasoning + Answer:

<think> To find $(r^2 + s^2)$ for the isosceles trapezoid with an inscribed circle, we need to use the properties of the trapezoid and the given information about its radius and area. Let's denote the trapezoid as $(ABCD)$ with $(AB = r)$, $(CD = s)$, and $(AD = BC = d)$ as the equal non-parallel sides. The radius of the inscribed circle is 3.

Since the trapezoid has an inscribed circle, the sum of the lengths of the parallel sides is equal to the sum of the lengths of the non-parallel sides. Therefore, we have:

$$r + s = 2d.$$

The area of the trapezoid can be expressed as:

$$\text{Area} = \frac{1}{2} (r + s) h,$$

where (h) is the height of the trapezoid. Given that the area is 72, we have:

$$72 = \frac{1}{2} (r + s) h.$$

Since $(r + s = 2d)$, we can substitute and get:

$$72 = d h.$$

Also, the height (h) can be related to the radius of the inscribed circle by the Pythagorean theorem in the right triangles formed by dropping perpendiculars from the points where the circle touches the non-parallel sides. The height (h) of the trapezoid is twice the radius, so:

$$h = 2 \times 3 = 6.$$

Substituting $(h = 6)$ into the area equation, we get:

$$72 = d \times 6,$$

which simplifies to:

$$d = 12.$$

Since $(r + s = 2d = 24)$, we need to find $(r^2 + s^2)$. We know that:

$$(r + s)^2 = r^2 + s^2 + 2rs.$$

From $(r + s = 24)$, we have:

$$24^2 = r^2 + s^2 + 2rs,$$
$$576 = r^2 + s^2 + 2rs.$$

We also know that the area can be expressed as:

$$72 = \frac{1}{2} (r + s) h = \frac{1}{2} \times 24 \times 6 = 72.$$

The semi-perimeter (p) of the trapezoid is:

$$p = \frac{r + s + 2d}{2} = \frac{24 + 24}{2} = 24.$$

The area (A) can also be expressed as:

$$A = rs + dh = rs + 12 \times 3 = rs + 36.$$

From the area, we know $(72 = 2rs)$, so $(rs = 36)$. Using this, we get:

$$576 = r^2 + s^2 + 2 \times 36,$$
$$576 = r^2 + s^2 + 72,$$
$$r^2 + s^2 = 504.$$

Thus, the value of $(r^2 + s^2)$ is:

$$\boxed{504}.$$

</think>
<answer>504</answer>